

3.4.3	Acids and Bases	
	<i>Brønsted–Lowry acid–base equilibria in aqueous solution</i>	know that an acid is a proton donor
		know that a base is a proton acceptor
		know that acid–base equilibria involve the transfer of protons
	<i>Definition and determination of pH</i>	know that $\text{pH} = -\log_{10}[\text{H}^+]$, where [] represents the concentration in mol dm^{-3}
		be able to convert concentration into pH and vice versa
		be able to calculate the pH of a solution of a strong acid from its concentration
	<i>The ionic product of water, K_w</i>	know that water is weakly dissociated
		know that $K_w = [\text{H}^+][\text{OH}^-]$
		be able to calculate the pH of a strong base from its concentration.
	<i>Weak acids and bases K_a for weak acids</i>	know that weak acids and weak bases dissociate only slightly in aqueous solution
		be able to construct an expression, with units, for the dissociation constant K_a for a weak acid
		know that $\text{p}K_a = -\log_{10} K_a$
		be able to perform calculations relating the pH of a weak acid to the dissociation constant, K_a , and the concentration
	<i>pH curves, titrations and indicators</i>	understand the typical shape of pH curves for acid–base titrations in all combinations of weak and strong monoprotic acids and bases
		be able to use pH curves to select an appropriate indicator
		be able to perform calculations for the titrations of monoprotic and diprotic acids with sodium hydroxide, based on experimental results
	<i>Buffer action</i>	be able to explain qualitatively the action of acidic and basic buffers
		know some applications of buffer solutions
		be able to calculate the pH of acidic buffer solutions